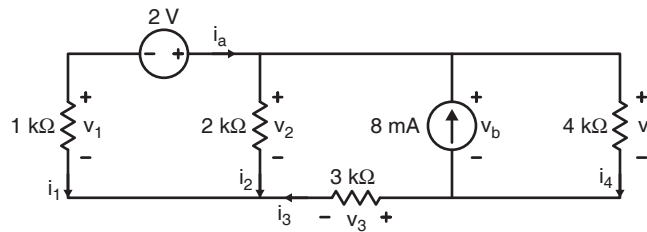
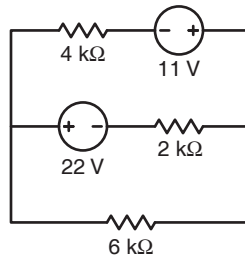


Issued: Mon Feb 03, 2025**Due:** Mon Feb 10, 2025**Problem 2.1**

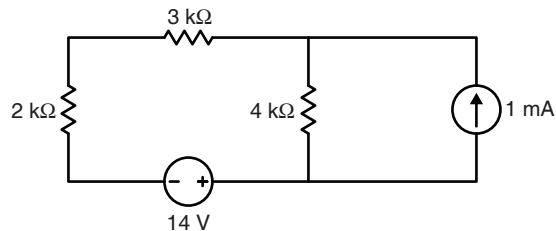
Consider the following circuit in which the unknown voltages and currents have been labelled for you.



- Determine the number of nodes n , the number of trivial nodes, the number of branches b and the number of sources s in this circuit.
- How many independent KCL relationships can you obtain for this circuit?
- Write down a set of independent KCL equations relating the currents in the circuit, and indicate which nodes are used in obtaining each equation.
- How many independent KVL relationships can you obtain for this circuit?
- Write down a set of independent KVL equations for the circuit, and indicate which loop or mesh is used for each equation.
- What are the terminal relations for the circuit?
- Solve the equations you derived in (c), (e), and (f) to determine all of the unknown currents and voltages shown in the figure. You may use Matlab, the MS Excel spreadsheet provided on the website, the online linear equation solver, or simply solve it yourself on paper. *Hint:* you should be able to eliminate 4 variables by applying the terminal relations and two more variables by using trivial nodes and trivial meshes.
- Calculate how much power is dissipated in each of the four resistors.
- Is the voltage source supplying or consuming power? How much power?
- Is the current source supplying or consuming power? How much power?

Problem 2.2

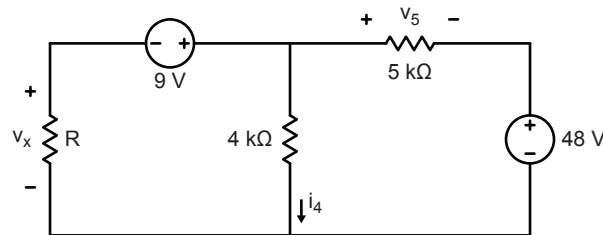
- (a) Solve for all of the unknown currents and voltages in this circuit. Please show all of your work, label all nodes and meshes that you use, and define current directions and polarities as needed. Use the passive sign convention for all resistors.
- (b) Calculate the power dissipated in each of the three resistors.
- (c) Calculate the power supplied (or consumed) by the 11 V source. Please indicate clearly whether the power is supplied or consumed.
- (d) Similarly, determine the power supplied (or consumed) by the 22 V source.

Problem 2.3

- (a) Solve for all of the unknown currents and voltages in this circuit. Please show all of your work, label all nodes and meshes that you use, and define current directions and polarities as needed. Use the passive sign convention for all resistors.
- (b) Calculate the power dissipated in each of the three resistors.
- (c) Calculate the power supplied (or consumed) by the voltage source. Please indicate clearly whether the power is supplied or consumed.
- (d) Similarly, determine the power supplied (or consumed) by the current source.

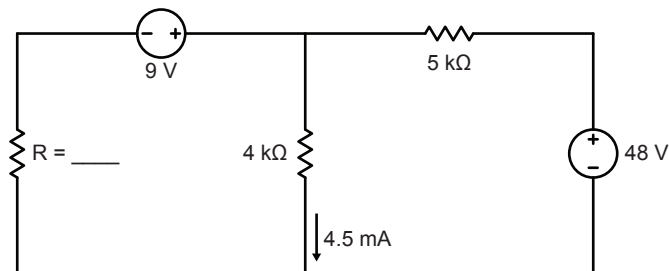
Problem 2.4

Note: parts (d)...(e) of this question can be answered independently of parts (a)...(c).



- Label all of the unknown branch voltages and currents in the above circuit (some of them are labeled for you.)
- Label all of the nodes on the circuit diagram above, using the letters $A, B, \dots$
- Find a set of *linearly independent* equations that *could* be used to solve for the unknown voltages and currents that you labeled in (a), (assuming that you knew the value of R .) Indicate clearly which node, loop, or resistor you are considering in each equation. You are not required to solve these equations.
- Suppose you perform a measurement and observe that $i_4 = 4.5 \text{ mA}$ (as shown below.)

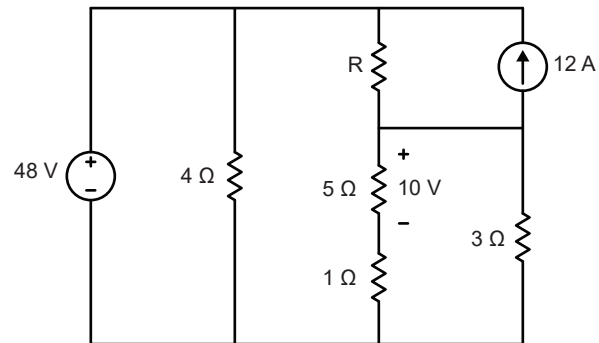
Find the remaining unknown voltages and currents, and determine the value of the unknown resistance R . Indicate your answers directly on the schematic at the bottom of this page. In all cases, show the actual direction of current flow and the true polarity of the voltages (i.e., none of your indicated quantities should be negative.)



- Complete the following statement:

The 9V source is [supplying/consuming] _____.
 (circle one) (amount of power)

Problem 2.5



In the above circuit, the voltage across the $5\ \text{k}\Omega$ resistor is observed to be 10 V, as shown. Use KVL, KCL and Ohms law to determine the value of the unknown resistance R . Use deduction – you should not need to write down or solve a large system of equations.